A trace inequality for four canonically normalized projections

Research proof and Lean formalization

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Abstract

We prove the four-projection inequality stated as Conjecture 6.6 in Banica–Nechita, arXiv:1602.04456v3. The canonical normalization uses the positive Moore–Penrose inverse square root of the sum of each pair of projections. The proof applies to arbitrary finite dimensions and does not require either of the two rank-balance assumptions in that conjecture. Its main step is an entrywise estimate in two independent spectral bases, followed by row and column bounds for rectangular contractions.

1 Statement and scope

For a positive semidefinite matrix T, write $T^{\dagger/2}$ for the positive operator with eigenvalue $t^{-1/2}$ at every positive eigenvalue t of T, and eigenvalue zero on its kernel. All traces below are ordinary, unnormalized traces. The displayed traces of products of orthogonal projections are real and nonnegative.

Theorem 1. *Let P, Q, R, S be orthogonal projections on a finite-dimensional complex Hilbert space. Suppose*

$$PQ = RS = 0, \qquad \operatorname{ran} P \cap \operatorname{ran} R = \operatorname{ran} Q \cap \operatorname{ran} S = \{0\}.$$

Put $D = (P+R)^{\dagger/2}$ and $E = (Q+S)^{\dagger/2}$, and define

$$P' = DPD, \quad R' = DRD, \quad Q' = EQE, \quad S' = ESE.$$

Then these four normalized operators are orthogonal projections, and

$$\operatorname{Tr}(P'Q') + \operatorname{Tr}(R'S') \le \operatorname{Tr}(PR) + \operatorname{Tr}(QS). \tag{1}$$

Conjecture 6.6 of [1, p. 20] additionally assumes $\operatorname{rank} P + \operatorname{rank} Q = \operatorname{rank} R + \operatorname{rank} S$ and $\operatorname{rank} P + \operatorname{rank} R = \operatorname{rank} Q + \operatorname{rank} S$. Neither equality is needed here. The source proves the case $S = 0$ in its Proposition 6.7. We make no claim about the general Sinkhorn-convergence questions elsewhere in that paper, strictness, or equality cases.

2 The scalar estimate

Lemma 2. *Let $-1 < a, b < 1$, and set $u = \sqrt{(1+a)(1+b)}$, $v = \sqrt{(1-a)(1-b)}$. For complex numbers k, l satisfying $uk + vl = 0$,*

$$|k+l|^2 \le (a^2+b^2)(|k|^2+|l|^2). \tag{2}$$

Proof. Both u and v are positive. Substitution of $l = -uk/v$ gives

$$|k+l|^2 = \left(1 - \frac{\sqrt{(1-a^2)(1-b^2)}}{1+ab}\right)(|k|^2 + |l|^2).$$

Write $t = a^2 + b^2$ and $w = \sqrt{(1-a^2)(1-b^2)}$. Since $1 + ab > 0$, it suffices to show $w \geq (1-t)(1+ab)$. This is immediate if $t \geq 1$. If $t < 1$ and $ab < 0$, then $w^2 = 1 - t + a^2b^2 \geq 1 - t \geq (1-t)^2$, so $w \geq 1 - t \geq (1-t)(1+ab)$. Finally, if $t < 1$ and $ab \geq 0$, the identity

$$(1-a^2)(1-b^2) - (1-a^2-b^2+ab)^2 = (a-b)^2(1-a^2-b^2)$$

implies $w \geq 1 - t + ab \geq (1-t)(1+ab)$. □

3 Frames, grading, and normalization

Choose orthonormal column frames p, r, q, s for the four ranges, and write

$$X = [p\ r], \quad Y = [q\ s], \qquad G = X^*X, \quad H = Y^*Y.$$

Both X and Y are injective. Indeed, $px + ry = 0$ puts $px = -ry$ in the intersection of the two ranges; disjointness and the isometry of each frame give $x = y = 0$. Thus G, H are positive definite on their respective coordinate spaces. Define

$$U = XG^{-1/2}, \quad V = YH^{-1/2}, \qquad K = U^*V.$$

Here $U^*U = I$ and $V^*V = I$; no equality between their column counts is used. In zero-dimensional coordinate spaces these statements have their usual empty-matrix meanings.

The precise ambient normalization is obtained from

$$(XX^*)^{\dagger/2}X = X(X^*X)^{-1/2} = U. \tag{3}$$

For completeness, a reduced singular-value decomposition $X = W\Sigma Z^*$, with positive diagonal Σ, gives the two sides as WZ^*; the ambient inverse square root is $W\Sigma^{-1}W^*$ and vanishes on $\mathrm{ran}(W)^\perp$. The empty-column case is immediate. Since $XX^* = P + R$, the columns of $U = [u_p\ u_r]$ are exactly $u_p = Dp$ and $u_r = Dr$. Consequently $P' = u_p u_p^*$ and $R' = u_r u_r^*$, with analogous formulas for Q', S'. The columns of U are orthonormal, so these are indeed orthogonal projections. In the corresponding block decomposition of K,

$$\mathrm{Tr}(P'Q') + \mathrm{Tr}(R'S') = \|K_{11}\|_{\mathrm{HS}}^2 + \|K_{22}\|_{\mathrm{HS}}^2. \tag{4}$$

This uses $\mathrm{Tr}(aa^*bb^*) = \|a^*b\|_{\mathrm{HS}}^2$.

Let $J = \mathrm{diag}(I, -I)$ on the (p, r) coordinates, and let L be its counterpart on the (q, s) coordinates. Put $A = G - I$, $B = H - I$. The isometry of each individual frame gives

$$A = \begin{pmatrix} 0 & p^*r \\ r^*p & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & q^*s \\ s^*q & 0 \end{pmatrix}, \qquad JAJ = -A, \quad LBL = -B.$$

In particular, $I - A = JGJ$ and $I - B = LHL$ are positive definite. The eigenvalues of A and B therefore belong to $(-1, 1)$. The assumptions $PQ = RS = 0$ give $p^*q = r^*s = 0$, and hence

$$J(X^*Y)L = -X^*Y.$$

Using $X = UG^{1/2}$, $Y = VH^{1/2}$ and the grading identities for the positive square roots, we obtain, with $K' = JKL$,

$$(I+A)^{1/2}K(I+B)^{1/2} + (I-A)^{1/2}K'(I-B)^{1/2} = 0. \tag{5}$$

4 Summation and conclusion

Diagonalize A and B in independent orthonormal bases, with eigenvalues a_i and b_j. Write k_{ij}, l_{ij} for the entries of K, K' in these bases. Equation (5) is exactly the hypothesis of Lemma 2 for each pair (i, j). Therefore

$$\begin{aligned} \tfrac{1}{4}\|K + K'\|_{\mathrm{HS}}^2 &\le \tfrac{1}{4}\sum_{i,j}(a_i^2 + b_j^2)(|k_{ij}|^2 + |l_{ij}|^2) \\ &\le \tfrac{1}{2}\left(\sum_i a_i^2 + \sum_j b_j^2\right). \end{aligned} \tag{6}$$

For the last step, $K = U^*V$ is a rectangular contraction because U, V are isometries; $K' = JKL$ is also a contraction. Thus every row and column of each matrix has squared norm at most one, including after the two independent unitary changes of basis. The last expression in (6) is $(\operatorname{Tr} A^2 + \operatorname{Tr} B^2)/2$. The off-diagonal block descriptions above give

$$\tfrac{1}{2}\operatorname{Tr} A^2 = \|p^*r\|_{\mathrm{HS}}^2 = \operatorname{Tr}(PR), \qquad \tfrac{1}{2}\operatorname{Tr} B^2 = \|q^*s\|_{\mathrm{HS}}^2 = \operatorname{Tr}(QS).$$

Finally, $(K + JKL)/2$ keeps precisely the two diagonal blocks of K. Combining this observation with (4) and (6) proves Theorem 1.

5 Certificate and research status

The accompanying development uses Lean 4.34.1 and mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

The development's two inequality endpoints are

`four_projection_inequality`
`banica_nechita_conjecture_6_6`.

The first omits both rank balances; the second adds exactly those source hypotheses. Both construct their range frames internally and use genuine continuous functional calculus for the ambient inverse square roots. The declaration `normalized_source_projection` proves the normalized operators are orthogonal projections. The certificate permits empty index types and has no residual frame, spectral or inverse-existence premise. Audited dependencies use only `propext`, `Classical.choice`, and `Quot.sound`. Independent integrated review and the final root extracted replay remain pending. Computational checks are supplementary.

The source statement used here is specifically arXiv version 3 of [1]. Comparison with the journal body remains incomplete. The documented bounded literature search has found no prior full resolution, but includes inaccessible material and does not establish worldwide priority. Polar decomposition, spectral calculus, and contraction estimates are standard tools. This draft remains unaccepted in the discovery campaign pending its stated gates.

References

[1] T. Banica and I. Nechita, *Flat matrix models for quantum permutation groups*, arXiv:1602.04456v3 (2016), Conjecture 6.6 and Proposition 6.7. Journal reference: Adv. Appl. Math. **83** (2017), 24–46, doi:10.1016/j.aam.2016.09.001.